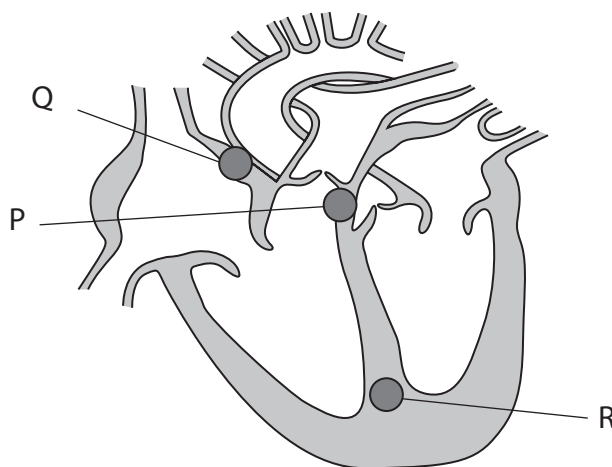


- 6 In humans the sinoatrial node, atrioventricular node and bundle of His are involved in regulation and coordination of the cardiac cycle.

(a) The diagram shows a human heart.



- (i) Which **one** of the following statements is correct?

(1)

- ☐ **A** Q is the sinoatrial node (SAN) and R is the atrioventricular node (AVN)
- ☐ **B** Q is the sinoatrial node (SAN) and P is the atrioventricular node (AVN)
- ☐ **C** Q is the bundle of His and R is the sinoatrial node (SAN)
- ☐ **D** R is the sinoatrial node (SAN) and P is the Purkinje fibres

- (ii) The electrical impulse that initiates the cardiac cycle begins without the need for a nerve impulse.

What is the term for this process?

(1)

- ☐ **A** diastole
- ☐ **B** myogenic
- ☐ **C** polarisation
- ☐ **D** systole

(b) A football player undertook regular exercise as part of a training programme.

During exercise there is a change in the duration of the cardiac cycle.

(i) Explain why there is a change in the cardiac cycle during this exercise.

(3)

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(ii) The table shows the effect of the training programme on the cardiac output and resting heart rate of this football player.

When measurements were made	Cardiac output / $\text{cm}^3 \text{ min}^{-1}$	Resting heart rate / bpm
Before training	4500	72
After training	4500	51

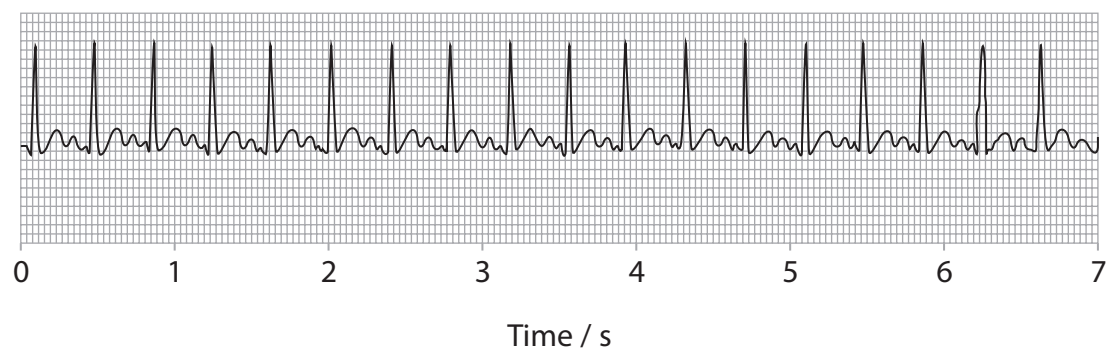
Calculate the stroke volume of this football player, after training.

(1)

Answer cm^3

(iii) After training, the footballer had an ECG.

This ECG trace is shown in the diagram.

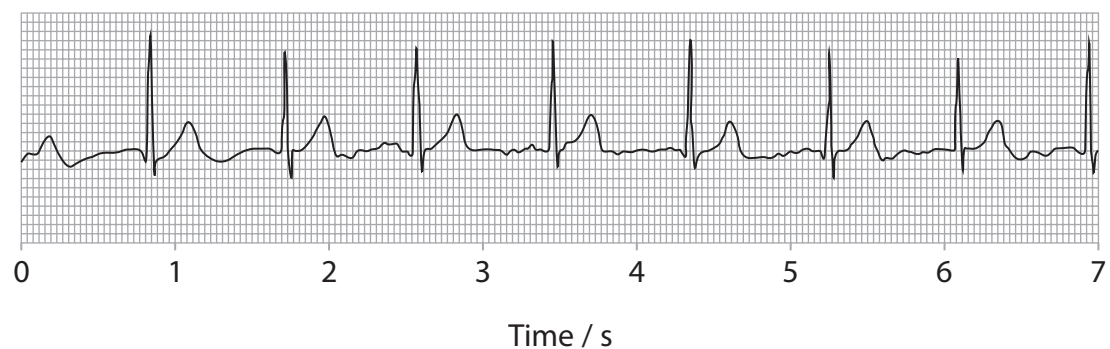


Calculate the heart rate of the footballer using the ECG in this diagram.

(1)

Answer

(iv) Two hours later a second ECG was recorded.



Comment on the changes that have occurred in the activity of the heart as shown in these two ECG traces.

(3)

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(Total for Question 6 = 10 marks)
